QQ-A-200/15B
30 November 1983
SUPERSEDING
QQ-A-200/15A
27 June 1974

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY BAR, ROD AND SHAPES, EXTRUDED, 7075-T76 (IMPROVED EXFOLIATION RESISTANT)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy 7075-T76 bar, rod and shapes, extruded described herein shall consist of this document and the latest issue of QQ-A-200/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 7075-T76 bar, rod and shapes produced by extrusion (see QQ-A-200/11 for specific requirements for 7075 extrusions in other tempers).

1.2.1 Tempers. The bar, rod and shapes shall be of the following tempers: T76, T76510 or T76511, as specified (see 6.2). The definition of these tempers shall be in accordance with QQ-A-200/GEN.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-200/GEN	-	Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for
QQ-A-200/11	-	Aluminum Alloy 7075, Bar, Rod, Shapes, Tube and Wire, Extruded

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

FSC 9525, 9530, 9540

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(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards

MIL-STD-2154 - Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

G34-72 - Exfoliation Corrosion Susceptibility in 7XXX Series Copper Containing Aluminum Alloys (EXCO Test)

G47 - Determining Susceptibility to Stress-Corrosion Cracking of High-Strength Aluminum Alloy Products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

3.2 Mechanical properties. Mechanical properties of as-supplied material in the longitudinal (extrusion) direction shall conform to the property requirements specified in table II, except as exempted from elongation requirements by QQ-A-200/GEN.

3.3 Internal defects. When specified (see 6.2), rod, bar and shapes shall be ultrasonically inspected (see QQ-A-200/GEN). Acceptance limits shall be as specified in table III. All extrusions in the inspection lot shall be inspected.

3.4 Resistance to exfoliation corrosion and stress-corrosion cracking.

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TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Copper	1.2	2.0
Magnesium	2.1	2.9
Manganese	-	0.30
Chromium	0.18	0.28
Titanium	-	0.20
Iron	-	0.50
Silicon	-	0.40
Zinc	5.1	6.1
Others, each	-	0.05
Others, total 2/	-	0.15
Aluminum	Remainder	

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits

2/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum

3.4.1 Exfoliation corrosion. Material tested in accordance with 4.2.4 shall not show exfoliation corrosion equivalent to or in excess of that depicted in figure 2 of ASTM G34-72 for category B. The frequency of tests shall be in accordance with QQ-A-200/GEN.

3.4.2 Stress-corrosion cracking. Extrusions of 0.750-inch thickness or thicker shall exhibit no susceptibility to stress-corrosion cracking under the conditions of the test specified in 4.2.5. The frequency of tests shall be in accordance with QQ-A-200/GEN.

3.4.3 Lot acceptance control criteria. Resistance to exfoliation corrosion and stress-corrosion cracking for each lot of material shall be determined by the following control criteria (see 4.1.2 and 4.2.6).

(a) Determine electrical conductivity and mechanical properties
(b) If the conductivity is 38.0 percent International Annealed Copper Standards (IACS) or higher, and the tensile properties meet the minimum limits specified herein, the material is acceptable
(c) If the conductivity is at least 36.0 percent IACS but less than 38.0 percent IACS, and the tensile properties meet the limits specified herein, the material shall either be tested as specified in 4.2.4 or be reprocessed
(d) If the conductivity is below 36.0 percent IACS, the material is not acceptable and must be reprocessed

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TABLE II. Mechanical properties

Temper	Specified thickness, (bar and shapes); diameter, rod, inches	Area, square inches	Ultimate tensile strength, minimum, ksi	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches or 4 times specimen diameter, minimum, percent
				minimum, ksi	Extension under load, inch per inch	
All (T76, T76510, and T76511)	up thru 0.049	All	73.0	63.0	0.0082	7
	0.050 - 0.124	All	74.0	64.0	0.0082	7
	0.125 - 0.249	up thru 20	74.0	64.0	0.0082	7
	0.250 - 0.499	up thru 20	75.0	65.0	0.0084	7
	0.500 - 1.000	up thru 20	75.0	65.0	0.0084	7
	1.001 - 2.000	up thru 20	75.0	65.0	0.0084	7
	2.001 - 3.000	up thru 20	74.0	64.0	0.0082	7
	3.001 - 4.000	up thru 20	74.0	63.0	0.0082	7

3.5 Marking of extrusions. In addition to marking as required in QQ-A-200/GEN, each extrusion shall be marked with its appropriate lot number in at least one location.

4. QUALITY ASSURANCE PROVISIONS (see QQ-A-200/GEN)

4.1 Sampling.

4.1.1 For chemical analysis, mechanical properties, and for exfoliation corrosion and stress-corrosion cracking tests. Samples for chemical analysis, determination of mechanical properties, exfoliation corrosion, and stress-corrosion cracking tests shall comply with the applicable requirements of QQ-A-200/GEN.

4.1.2 For conductivity tests. Conductivity shall be measured on the same specimens used for mechanical property tests. Conductivity shall be measured prior to testing for mechanical properties.

4.2 Test methods.

4.2.1 Chemical analysis. Samples selected in accordance with 4.1.1. shall be analyzed to determine conformance to 3.1 using the test methods specified in QQ-A-200/GEN.

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TABLE III. Ultrasonic discontinuity acceptance limits 1/, 2/

Thickness (bar and shapes), inches 3/	Maximum weight of piece (pounds)	Maximum width to thickness ratio	Discontinuity class 4/
0.500 - 1.499	600	10 to 1	B
1.500 and over	600	10 to 1	A

1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas
2/ Requirements for ultrasonic inspection for sizes, maximum weights and discontinuity classes differing from those shown in table III and for rod shall be specified in the contract provided the techniques and standards are agreed to by the procuring activity and the contractor (see 6.2)
3/ For thickness of a shape, see definition for shape in QQ-A-200/GEN
4/ The discontinuity class limits are defined in MIL-STD-2154

4.2.2 Mechanical properties. Samples selected in accordance with 4.1.1 shall be prepared for testing and tested in accordance with QQ-A-200/GEN to determine conformance to 3.2.

4.2.3 Ultrasonic inspection. When ultrasonic inspection is specified (see 6.2), extrusions shall be inspected in accordance with MIL-STD-2154 to determine conformance to 3.3.

4.2.4 Exfoliation corrosion test. Samples taken in accordance with 4.1.1 shall be tested for conformance to 3.4.1 using the test method specified by ASTM G34-72. Full-size extrusions or sections of extrusions, sufficiently large to provide visual assessment of attack, shall be used. For all test coupons, 10 percent of the thickness shall be removed by machining one surface. The machined surface shall be the surface evaluated.

4.2.5 Stress-Corrosion Cracking Test. In order to determine conformance to 3.4.2, stress-corrosion cracking tests shall be performed in accordance with ASTM G47. Specimens shall be stressed in the short transverse direction with respect to grain flow and held at constant strain. The stress level shall be 25 ksi, and the test duration shall be 20 days.

4.2.6 Electrical conductivity tests. Specimens selected in accordance with 4.1.2 shall be tested for conformance to 3.4.3. Specimen preparation and testing shall be in accordance with QQ-A-200/GEN.

4.3 Test records. It is the responsibility of the contractor that all test results be recorded, identified as to lot, and preserved after completion of the contract or purchase order. These records shall be made available to the Federal Government upon request.

5. PREPARATION FOR DELIVERY (see QQ-A-200/GEN)

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6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required, and where exfoliation resistance superior to that of 7075-T6 is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Form and temper required (see 1.2)
(c) Dimensions required
(d) Requirements for sizes not specifically covered (see QQ-A-200/GEN)
(e) Special end use requirements
(f) Whether ultrasonic inspection of extrusions is required (see 3.3)
(g) Selection of applicable levels of preservation, packaging, and packing required, whether level A, level B, or commercial (see QQ-A-200/GEN)

MILITARY INTEREST:

Custodians

Army-MR
Navy-AS
Air Force-20

Review Activities

Army-AV
Navy-SH
Air Force-99

User Activities

Army-EL
Navy-EC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
NASA-JFK
DOE-BPA

PREPARING ACTIVITY:

NAVY-AS

DoD Project 9525-0065

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specificaiton to obtain extra copies and other documents referenced herein.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
QQ-A-200/15B	Aluminum Alloy Bar, Rod and Shapes, Extruded, 7075-T76 (Improved Exfoliation Resistant)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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